

Research Notes on Mathematical optimization

Mathematical optimization

>> Section 1

argmin $x^2 + 1$, subject to: $x \in (-\infty, -1]$.
$$\underset{x}{\operatorname{argmin}} x^2 + 1, \text{subject to: } x \in (-\infty, -1].$$

>> Section 2

Given: a function $f: A \rightarrow \mathbb{R}$ from some set A to the real numbers Sought: an element $x_0 \in A$ such that $f(x_0) \leq f(x)$ for all $x \in A$ ("minimization") or such that $f(x_0) \geq f(x)$ for all $x \in A$ ("maximization"). Such a formulation is called an optimization problem or a mathematical programming problem (a term not directly related to computer programming, but still in use for example in linear programming – see History below). Many real-world and theoretical problems may be modeled in this general framework. Since the following is valid:

>> Section 3

This represents the value (or values) of the argument x in the interval $(-\infty, -1]$ that minimizes (or minimize) the objective function $x^2 + 1$ (the actual minimum value of that function is not what the problem asks for). In this case, the answer is $x = -1$, since $x = 0$ is infeasible, that is, it does not belong to the feasible set. Similarly,

>> Section 4

Sufficient conditions for optimality While the first derivative test identifies points that might be extrema, this test does not distinguish a point that is a minimum from one that is a maximum or one that is neither. When the objective function is twice differentiable, these cases can be distinguished by checking the second derivative or the matrix of second derivatives (called the Hessian matrix) in unconstrained problems, or the matrix of second derivatives of the objective function and the constraints called the bordered Hessian in constrained problems. The conditions that distinguish maxima, or minima, from other stationary points are called 'second-order conditions' (see 'Second derivative test'). If a candidate solution satisfies the first-order conditions, then the satisfaction of the second-order conditions as well is sufficient to establish at least local optimality.

>> Section 5

$f(x_0) \geq f(x) \Leftrightarrow -f(x_0) \leq -f(x)$,
$$f(\mathbf{x}_0) \geq f(\mathbf{x}) \Leftrightarrow -f(\mathbf{x}_0) \leq -f(\mathbf{x}),$$

>> Section 6

Adding more than one objective to an optimization problem adds complexity. For example, to optimize a structural

design, one would desire a design that is both light and rigid. When two objectives conflict, a trade-off must be created. There may be one lightest design, one stiffest design, and an infinite number of designs that are some compromise of weight and rigidity. The set of trade-off designs that improve upon one criterion at the expense of another is known as the Pareto set. The curve created plotting weight against stiffness of the best designs is known as the Pareto frontier. A design is judged to be "Pareto optimal" (equivalently, "Pareto efficient" or in the Pareto set) if it is not dominated by any other design: If it is worse than another design in some respects and no better in any respect, then it is dominated and is not Pareto optimal. The choice among "Pareto optimal" solutions to determine the "favorite solution" is delegated to the decision maker. In other words, defining the problem as multi-objective optimization signals that some information is missing: desirable objectives are given but combinations of them are not rated relative to each other. In some cases, the missing information can be derived by interactive sessions with the decision maker. Multi-objective optimization problems have been generalized further into vector optimization problems where the (partial) ordering is no longer given by the Pareto ordering.

>> Section 7

The iterative methods used to solve problems of nonlinear programming differ according to whether they evaluate Hessians, gradients, or only function values. While evaluating Hessians (H) and gradients (G) improves the rate of convergence, for functions for which these quantities exist and vary sufficiently smoothly, such evaluations increase the computational complexity (or computational cost) of each iteration. In some cases, the computational complexity may be excessively high. One major criterion for optimizers is just the number of required function evaluations as this often is already a large computational effort, usually much more effort than within the optimizer itself, which mainly has to operate over the N variables. The derivatives provide detailed information for such optimizers, but are even harder to calculate, e.g. approximating the gradient takes at least $N+1$ function evaluations. For approximations of the 2nd derivatives (collected in the Hessian matrix), the number of function evaluations is in the order of N^2 . Newton's method requires the 2nd-order derivatives, so for each iteration, the number of function calls is in the order of N^2 , but for a simpler pure gradient optimizer it is only N . However, gradient optimizers need usually more iterations than Newton's algorithm. Which one is best with respect to the number of function calls depends on the problem itself.

>> Section 8

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asks for the maximum value of the objective function $2x$, where x may be any real number. In this case, there is no such maximum as the objective function is unbounded, so the answer is "infinity" or "undefined".

>> Section 9

An optimization problem with discrete variables is known as a discrete optimization, in which an object such as an integer, permutation or graph must be found from a countable set. A problem with continuous variables is known as a continuous optimization, in which optimal arguments from a continuous set must be found. They can include constrained problems and multimodal problems. An optimization problem can be represented in the following way:

>> Section 10

Geophysics Optimization techniques are regularly used in geophysical parameter estimation problems. Given a set of geophysical measurements, e.g. seismic recordings, it is common to solve for the physical properties and geometrical shapes of the underlying rocks and fluids. The majority of problems in geophysics are nonlinear with both deterministic and stochastic methods being widely used.

>> Section 11

Optimization techniques are used in many facets of computational systems biology such as model building, optimal experimental design, metabolic engineering, and synthetic biology. Linear programming has been applied to calculate the maximal possible yields of fermentation products, and to infer gene regulatory networks from multiple microarray datasets as well as transcriptional regulatory networks from high-throughput data. Nonlinear programming has been used to analyze energy metabolism and has been applied to metabolic engineering and parameter estimation in biochemical pathways.